HDOC DAY 71

PROBLEM

Write a C program to dynamically allocate two matrices and perform addition, subtraction, and multiplication using pointer notation. Implement each operation as a separate function and release all allocated memory properly.

CODE

```
#include <stdio.h>

#include <stdlib.h>


int **createMatrix(int rows, int cols)

{

   int **mat = malloc(rows * sizeof(int *));


   for (int i = 0; i < rows; i++)

      *(mat + i) = malloc(cols * sizeof(int));


   return mat;

}


void inputMatrix(int **mat, int rows, int cols)

{

   for (int i = 0; i < rows; i++)

      for (int j = 0; j < cols; j++)

         scanf("%d", (*(mat + i) + j));

}
```

```
void displayMatrix(int **mat, int rows, int cols)
{
   for (int i = 0; i < rows; i++)
   {
      for (int j = 0; j < cols; j++)
         printf("%d ", *(*(mat + i) + j));

      printf("\n");
   }
}

void addMatrix(int **a, int **b, int **result, int rows, int cols)
{
   for (int i = 0; i < rows; i++)
      for (int j = 0; j < cols; j++)
         *(*(result + i) + j) =
            *(*(a + i) + j) + *(*(b + i) + j);
}

void subtractMatrix(int **a, int **b, int **result, int rows, int cols)
{
   for (int i = 0; i < rows; i++)
      for (int j = 0; j < cols; j++)
         *(*(result + i) + j) =
            *(*(a + i) + j) - *(*(b + i) + j);
}

void multiplyMatrix(int **a, int **b, int **result,
```

```
                int r1, int c1, int c2)
{
   for (int i = 0; i < r1; i++)
   {
      for (int j = 0; j < c2; j++)
      {
         *(*(result + i) + j) = 0;

         for (int k = 0; k < c1; k++)
         {
            *(*(result + i) + j) +=
               *(*(a + i) + k) * *(*(b + k) + j);
         }
      }
   }
}

void freeMatrix(int **mat, int rows)
{
   for (int i = 0; i < rows; i++)
      free(*(mat + i));

   free(mat);
}

int main()
{
   int r1, c1, r2, c2;
```

```
printf("Enter rows and columns of Matrix A: ");
scanf("%d %d", &r1, &c1);

printf("Enter rows and columns of Matrix B: ");
scanf("%d %d", &r2, &c2);

int **a = createMatrix(r1, c1);
int **b = createMatrix(r2, c2);

printf("Enter elements of Matrix A:\n");
inputMatrix(a, r1, c1);

printf("Enter elements of Matrix B:\n");
inputMatrix(b, r2, c2);

/* Addition and Subtraction */
if (r1 == r2 && c1 == c2)
{
   int **result = createMatrix(r1, c1);

   addMatrix(a, b, result, r1, c1);

   printf("\nAddition:\n");
   displayMatrix(result, r1, c1);

   subtractMatrix(a, b, result, r1, c1);
```

```
    printf("\nSubtraction:\n");
    displayMatrix(result, r1, c1);

    freeMatrix(result, r1);
}
else
{
    printf("\nAddition and subtraction are not possible.\n");
}

/* Multiplication */
if (c1 == r2)
{
    int **result = createMatrix(r1, c2);

    multiplyMatrix(a, b, result, r1, c1, c2);

    printf("\nMultiplication:\n");
    displayMatrix(result, r1, c2);

    freeMatrix(result, r1);
}
else
{
    printf("\nMultiplication is not possible.\n");
}

freeMatrix(a, r1);
```

```
    freeMatrix(b, r2);


    return 0;
}
```